

Assignment 4

Due by 11:59 pm on 29 November 2025

Covered contents:

Lectures 10-12

Submission method:You should submit a **single** PDF file for all written answers.**A note on using generative tools:**

You are welcome to use generative tools to assist you in completing assignments. However, you are not allowed to simply copy and paste; you should understand the outputs and submit works that are completed based on your own understanding. Failure to do so may result in penalties.

If you have any questions, please email phycsan@ust.hk.

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Question 1: Practicing with Quantum Systems [20 points]

- (a) Consider a one-qubit system prepared in the state

$$|\psi\rangle = N(2|0\rangle - |1\rangle). \quad (1)$$

- (i) Assuming N is real, find N .
(ii) Suppose we measure an observable A given by the operator

$$A = \lambda|0\rangle\langle 0| + \epsilon|0\rangle\langle 1| + \epsilon|1\rangle\langle 0| + \lambda|1\rangle\langle 1|. \quad (2)$$

What are the necessary conditions we must impose on λ and ϵ ?

- (iii) Find the eigenvalues of A in terms of λ and ϵ . What are the corresponding probabilities of measuring them?
- (b) Consider a two-qubit system in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \quad (3)$$

Show that this state cannot be written as

$$|\psi\rangle = |\phi_0\rangle \otimes |\phi_1\rangle, \quad (4)$$

where $|\phi_0\rangle$ and $|\phi_1\rangle$ are one-qubit states. In physics parlance, we say that the system is “entangled”.

Here’s one way to prove it: Expand $|\psi\rangle$ in the computational basis, and figure out constraints on the coefficients.

- (c) Generalize your solution in (b) to the following situation:

Consider an n -dimensional Hilbert space $\mathcal{H}$ with basis $\{|e_i\rangle\}_{i=1}^n$. Given an arbitrary state $|\psi\rangle$ in $\mathcal{H} \otimes \mathcal{H}$, we may write it as

$$|\psi\rangle = \sum_{i,j} \psi_{ij} |e_i\rangle |e_j\rangle. \quad (5)$$

Suppose we interpret the coefficients ψ_{ij} as an $n \times n$ matrix, what can you say about this matrix’s determinant if the state $|\psi\rangle$ is “unentangled” (i.e. it can be written as a product state)?

The next question (part (d)) is optional. It does not count towards your grades.

- (d) Given your answer in (c), how likely is it that a random state in $\mathcal{H} \otimes \mathcal{H}$ is entangled? (Answering this rigorously requires some advanced maths. For us, it suffices to do a bit of hand-waving.)

Question 2: Quantum Circuits [20 points]

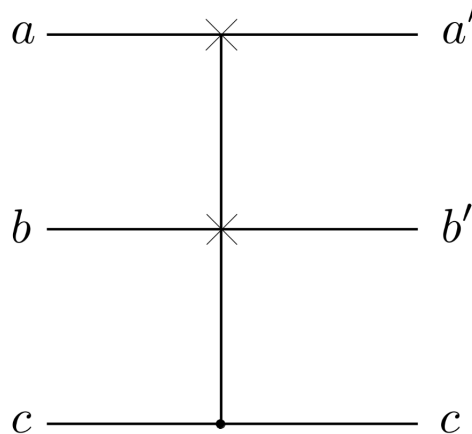


Figure 1: Circuit representation of the Fredkin gate, where the third qubit is used as the control qubit.

- (a) The Fredkin gate is defined as a controlled SWAP gate: it accepts three qubits, and swap two of the input qubits if the third control qubit is set to be 1, otherwise it does nothing.
- Write down Fredkin gate's action on the 8 computational basis states. Use the third qubit as the control qubit.
 - Express Fredkin gate as an 8-by-8 matrix in the computational basis. Order the basis states in increasing lexicographic order.
 - Show and prove how to use Fredkin gate to reproduce a classical AND gate. Note that classically the AND gate is non-reversible!
 - Show and prove how to construct a Fredkin gate if you are only allowed to use Toffoli gates.
- (b) Construct a quantum circuit that will perform the following operations:

$$\begin{aligned} |00\rangle &\mapsto \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \\ |01\rangle &\mapsto \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \\ |10\rangle &\mapsto \frac{|00\rangle - |11\rangle}{\sqrt{2}}, \\ |11\rangle &\mapsto \frac{|01\rangle - |10\rangle}{\sqrt{2}}. \end{aligned}$$

It turns out this circuit is very useful in various applications (e.g. quantum communications)!

Hint: These operations should remind you of two particular gates we mentioned extensively during lecture...

Question 3: Analysis of Grover's Algorithm [10 points]

During lecture, we showed how Grover's algorithm is essentially just the repeated application of the quantum oracle and the Grover operator G defined as

$$G = 2|b\rangle\langle b| - \mathbb{1},$$

where $|b\rangle$ is the 'balanced' state.

We also analyzed the correctness of this algorithm in the case where there is only one solution to the search problem. It turns out we can generalize our analysis to the more general case where there are multiple, say M , solutions.

Let us define the following two (normalized) states:

$$|\alpha\rangle = \frac{1}{\sqrt{N-M}} \sum_{x \text{ is not a solution}} |x\rangle, \\ |\beta\rangle = \frac{1}{\sqrt{M}} \sum_{x \text{ is a solution}} |x\rangle.$$

- (a) Consider an arbitrary state $|\psi\rangle$ in the subspace spanned by $|\alpha\rangle$ and $|\beta\rangle$. Show that the quantum oracle $\mathcal{O}_f$ and the Grover operator G do not take $|\psi\rangle$ out of this subspace.
- (b) Given your result in (a), our geometrical analysis of Grover's algorithm carries through similarly in this more general case, except now the final state vector should point in the $|\beta\rangle$ direction. However, the time complexity is no longer $\mathcal{O}(\sqrt{N})$. What should be the correct time complexity? Show your steps clearly.